

Nature Technical Report: Next-Generation MR Thermometry

Quantum Restricted Boltzmann Machines, Bayesian Distributions, and Qubit Photonic Coil Topology

ABSTRACT

This report outlines the theoretical foundation and finite mathematical derivations of two novel Magnetic Resonance (MR) Thermometry sequences: Quantum Restricted Boltzmann Machine (RBM) Thermometry and Statistical Bayesian Distribution Thermometry. Furthermore, we formalize the topological routing pathways for two discrete neurovascular imaging elements, termed 'Qubit Photonic Head Coils', which leverage deterministic phase interleaving (bit-shuffling) to attain high SNR in continuous data acquisition routines.

1. Quantum RBM Thermometry

MR Thermometry using Proton Resonance Frequency (PRF) Shift depends heavily on phase mapping susceptibilities. The Quantum Restricted Boltzmann Machine (RBM) attempts to denoise PRF inferences through simulated energy minimization of the MR signal manifold.

1.1 Finite Energy Function

The visible units v_i correlate to the finite Cartesian-sampled k-space signal, and h_j the sub-voxel latent thermal characteristics.

$$E(v, h) = - \sum_i a_i v_i - \sum_j b_j h_j - \sum_{\{i,j\}} v_i w_{\{ij\}} h_j$$

(Eq. 1 — RBM Energy)

1.2 Phase Shift Modulator

A proxy for real-world MR interaction is established. Finite approximation yields local phase alterations on reconstructed magnitude matrices.

$$\Delta\Phi_{\text{thermal}} \approx \gamma_H \cdot \alpha_{\text{prf}} \cdot B^{\blacksquare} \cdot TE \cdot \nabla T$$

(Eq. 2 — PRF Phase Shift)

2. Statistical Bayesian Thermometry

By enforcing a Gaussian prior on finite structural edges, we obtain localized spatial homogeneity priors combined with the posterior thermal likelihoods.

2.1 Gradient Probability Map

Thus creating a spatial temperature estimator heavily penalized for noisy thermal distributions outside physical boundaries.

$$P(\Theta|M) \propto \exp[- ||\nabla M(x,y)||^2 / (2\sigma_{\text{edge}}^2)] \cdot \Pi_{\{x,y\}} N(T_{\text{est}} | \mu(x,y), \sigma_{\text{obs}}^2)$$

(Eq. 3 — Bayesian Posterior)

3. Qubit Photonic Neurovascular Array Coils

Hardware limitations in standard volumetric coil transmission can be overcome by mimicking qubit state logic and algorithmic signal routing. This entails bit-reversal and discrete shuffling schemes applied spatially.

3.1 V1: Strict Bit-Reversal Shuffling

For a 32-element coil (5-bit state representation), contiguous elements often alias adjacent noise features. By reversing the binary representation, element arrays map to physically disparate sectors:

$$\text{Shuffle}(n) = (n_0, n_1, n_2, n_3, n_4) \rightarrow (n_4, n_3, n_2, n_1, n_0)_2$$

(Eq. 4 — Bit Reversal)

$$\phi_k = \exp[i(\pi/4) \cdot \text{Shuffle}(k)]$$

(Eq. 5 — Element Phase)

3.2 V2: Depth-Tuned Interleaving Shuffling

For deep cortical penetration in a 64-element matrix, alternating bits dictates radial positioning across double-tiered geometry.

$$\begin{aligned} R(n) &= R_{\text{min}} - \Delta r \cdot (n \bmod 2) \\ \text{Interleave}(n) &= U_{\{m=0\}}^{^2} (n_{\{2m+1\}}, n_{\{2m\}}) \end{aligned}$$

(Eq. 6 — Radially Interleaved Geometry)

Yielding distinct spatial phases capable of constructively interfering over specific cerebrovascular landmarks.